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Education

University of Colorado Boulder <i>Ph.D. in Mathematics</i> <i>Advised by Agnès Beaudry</i>	May 2025
University of Texas Austin <i>B.Sc. in Physics</i>	Dec. 2018
University of Texas Austin <i>B.Sc. in Mathematics</i>	May 2018

Research Interests

My research interests lie generally in algebraic topology and stable homotopy theory. My work currently focuses on the intersection of stable equivariant and chromatic homotopy theory. I am also interested in interactions between algebraic topology and physics.

Publications

- [1] **Apr. 2026.** “Duals of higher real K -theories at $p = 2$ ”. In: *Advances in Mathematics* 495, p. 110974. ISSN: 0001-8708. DOI: <https://doi.org/10.1016/j.aim.2026.110974> [🔗](#). URL: <https://www.sciencedirect.com/science/article/pii/S0001870826001969>.
- [2] **Mar. 2024** (with Agnès Beaudry, Michael Hermele, Markus J. Pflaum, Marvin Qi, and Daniel D. Spiegel). “Homotopical foundations of parametrized quantum spin systems”. In: *Reviews in Mathematical Physics*. ISSN: 1793-6659. DOI: [10.1142/s0129055x24600031](https://doi.org/10.1142/s0129055x24600031) [🔗](#).
- [3] **Sept. 2023** (with Xueda Wen, Marvin Qi, Agnès Beaudry, Markus J. Pflaum, Daniel Spiegel, Ashvin Vishwanath, and Michael Hermele). “Flow of higher Berry curvature and bulk-boundary correspondence in parametrized quantum systems”. In: *Physical Review B* 108.12, p. 125147. ISSN: 2469-9969. DOI: [10.1103/physrevb.108.125147](https://doi.org/10.1103/physrevb.108.125147) [🔗](#).
- [4] **Aug. 2022** (with Daniel Spiegel, Marvin Qi, Michael Hermele, Agnès Beaudry, and Markus J. Pflaum). “Continuous dependence on the initial data in the Kadison transitivity theorem and GNS construction”. In: *Reviews in Mathematical Physics* 34.09. ISSN: 1793-6659. DOI: [10.1142/s0129055x22500313](https://doi.org/10.1142/s0129055x22500313) [🔗](#).

Preprints

- [5] **Aug. 2026.** “On homological Real trace methods”. In: *arXiv e-prints*, arXiv:2608.04297, arXiv:2608.04297. DOI: [10.48550/arXiv.2608.04297](https://doi.org/10.48550/arXiv.2608.04297) [🔗](#). arXiv: [2608.04297](https://arxiv.org/abs/2608.04297) [\[math.AT\]](#) [🔗](#).

Invited Talks

* = upcoming

1. UIUC Topology Seminar Urbana Champagne, IL, US.	Nov. 2026*
2. Midwest Topology Seminar East Lansing, MI, US.	May 9, 2026
3. University of Kentucky Topology seminar Lexington, KY, US.	Apr. 9, 2026
4. JMM: AMS special session on <i>homotopy theory</i> Seattle, WA, US.	Jan. 11, 2025

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| 5. | University of Virginia Topology seminar
Charlottesville, VA, US. | Nov. 14, 2024 |
| 6. | Michigan State University Topology seminar
East Lansing, MI, US. | Nov. 5, 2024 |
| 7. | UI Bloomington Topology seminar
Bloomington, IN. US. | Oct. 9, 2024 |
| 8. | Algebraic Structures in Topology II
San Juan, PR, US. | June 6, 2024 |
| 9. | UI Urbana Champaign Topology seminar
Champaign Urbana, IL, US. | Apr. 30, 2024 |
| 10. | UCLA Algebraic Topology seminar
Los Angeles, CA, US. | Jan. 10, 2024 |
| 11. | JMM: AIM special session on <i>equivariant techniques in stable homotopy theory</i>
San Francisco, CA, US. | Jan. 6, 2024 |
| 12. | JMM: AMS special session on <i>homotopy theory</i>
San Francisco, CA, US. | Jan. 4, 2024 |

Selected Talks

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| 1. | “The C_2 -equivariant cohomology of C_2 -equivariant Eilenberg-MacLane spaces”
eCHT Hopf Rings Seminar. Online. | Mar. 2024 |
| 2. | “Duality for higher real K -theory spectra”
Groot Summer seminar. Online. | Aug. 2023 |
| 3. | “Chromatic homotopy theory and duality”
CU Boulder Topology seminar. Boulder, CO, US. | Oct. 2022 |
| 4. | “Postnikov invariants of topological phases of matter”
CU Boulder Grad Student seminar. Boulder, CO, US. | Aug. 2022 |
| 5. | “Extended Positivity and Stable Homotopy Theory”
Seminar on Reflection Positivity and Invertible Phases. Online. | Apr. 2022 |
| 6. | “TQFTs from fusion categories”
CU Boulder Topology Seminar. Boulder, CO, US. | Mar. 2022 |
| 7. | “Clifford Modules”
eCHT Kan Seminar. Online. | Feb. 2022 |
| 8. | “Bordism & stable homotopy theory”
CU Boulder Grad Student seminar. Boulder, CO, US. | Mar. 2021 |
| 9. | “Atiyah duality”
eCHT Kan Seminar. Online. | Oct. 2020 |
| 10. | “Complex-oriented cohomology theories”
CU Boulder Topology Seminar. Boulder, CO, US. | Oct. 2019 |
| 11. | “de Rham Cohomology”
UT Mathematics Undergraduate Student Talks. Austin, TX, US. | May 2018 |
| 12. | “Klein geometry”
UT Mathematics Undergraduate Student Talks. Austin, TX, US. | Dec. 2017 |

Seminars Organized

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| 1. CU Boulder graduate student seminar on <i>Category Theory</i>
Boulder, CO, US. | Fall 2022
Fall 2021 |
| 2. CU Boulder seminar on <i>Modular Tensor Categories</i>
Boulder, CO, US. | Spring 2022 |

REU/Gs

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| 1. CU Boulder REU/G on <i>Topology and lattice models (Mentor)</i>
Boulder, CO, US. | Summer 2022 |
| 2. CU Boulder REU/G on <i>Equivariant cohomology of topological spaces</i>
Online. | Summer 2020 |

Awards & Fellowships

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| 1. Richard Laver Graduate Fellowship | Summer 2024 |
| 2. Graduate School Travel Grant | Spring 2024 |
| 3. Leonhardy Summer Fellowship | Summer 2023 |
| 4. CU Boulder Graduate School Fellowship | Summer 2022 |
| 5. Haller Summer Fellowship | Summer 2021 |

Teaching Experience

* = upcoming

University of Notre Dame

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| 1. MATH 20580 Intro to Linear Algebra & Differential Equations | Spring 2026 |
| 2. MATH 10350 Calc A | Fall 2025 |

University of Colorado Boulder: Instructor

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| 1. MATH 2300 Calc. 2 | Spring 2025 |
| 2. MATH 2400 Calc. 3 | Fall 2022 |
| 3. MATH 2300 Calc. 2 | Fall 2021 |

University of Colorado Boulder: Teaching Assistant

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| 1. MATH 2300 Calc. 2 | Spring 2024 |
| 2. MATH 2300 Calc. 2
MATH 2400 Calc. 3 | Spring 2023 |
| 3. MATH 2400 Calc. 3 | Spring 2021 |
| 4. MATH 2300 Calc. 2 | Spring 2020 |

University of Texas at Austin: Learning Assistant

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| 1. M408S Integral Calculus for Science | Spring 2018 |
| 2. M408L Integral Calculus | Fall 2017 |

Service

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| 1. CU Boulder diversity committee
Organized department trainings. | Aug. 2024 - present |
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| 2. | CU Boulder prelim mentor
Mentored math grad students preparing for preliminary exams. | July - Aug. 2023 |
| 3. | CU Boulder first-year graduate student mentor
Mentored first-year math graduate students. | Aug. 2021-May 2022 |
| 4. | CU Boulder AMS member at large
Coordinated activities related to preliminary exams. | Aug. 2020-May 2021 |

Conferences & Workshops

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| 1. | BIRS Workshop on “Equivariant homotopy and algebraic K -theory”
Banff, Alberta, CA. | Feb. 21-26, 2027* |
| 2. | Midwest Topology Seminar
East Lansing, MI, US. | May 9-10, 2026* |
| 3. | AIM Workshop on <i>Computations in stable homotopy theory</i>
Pasadena, CA, US. | Oct. 27-31, 2025 |
| 4. | Scissors Congruence and K-theory Conference
Philadelphia, PA, US. | July 14-17, 2025 |
| 5. | INI Programme: <i>Equivariant homotopy theory in context</i>
Cambridge, UK. | June. 2025 |
| 6. | AMS Joint Mathematics Meetings
Seattle, WA, US. | Jan. 10-11 2025 |
| 7. | SLMath Workshop on <i>Hot Topics: Life after the Telescope conjecture</i>
Berkley, CA, US. | Dec. 8 - 13, 2024 |
| 8. | AIM Workshop on <i>Chromatic homotopy and p-adic geometry</i>
Pasadena, CA, US. | Dec. 2 - 6, 2024 |
| 9. | Talbot Workshop on <i>Topological Cyclic Homology of Ring Spectra</i>
Nacogdoches, TX, US. | Aug. 11 - 17, 2024 |
| 10. | Algebraic K -theory Collaborative Workshop
Nashville, TN, US. | July 29 - Aug.
2, 2024 |
| 11. | Algebraic Structures in Topology II
San Juan, PR, US. | June 4 - 14, 2024 |
| 12. | eCHT research workshop on Hopf rings
Online. | June 24 - 28, 2024 |
| 13. | Midwest Topology Seminar
Bloomington, IN, US. | Apr. 27, 2024 |
| 14. | AMS Joint Mathematics Meetings
San Fransisco, CA, US. | Jan. 3 - 6, 2024 |
| 15. | South Central Topology Conference III
Las Cruces, NM, US. | Oct. 28 - 29, 2023 |
| 16. | Homotopy theory in honor of Paul Goerss
Evanston, IL, US. | Mar. 20 - 24, 2023 |
| 17. | AIM Workshop on <i>Equivariant techniques in stable homotopy theory</i>
San Jose, CA, US. | Oct. 24 - 28, 2022 |
| 18. | Homotopical methods in fixed point theory
Boulder, CO, US. | July 11 - 15, 2022 |

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| 19. | Algebraic Structures in Topology
San Juan, PR, US. | May 30-June 3, 2022 |
| 20. | Cornell Topology Festival
Ithaca, NY, US. | May 6 - 8, 2022 |
| 21. | Viva Talbot
Online. | June 1 - 6, 2021 |
| 22. | AMS Joint Mathematics Meetings
Denver, CO, US. | Jan. 15 - 18, 2020 |

Selected Seminars

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| 1. | CU Boulder Topology seminar | Fall 2019 - present |
| 2. | CU Boulder Math physics seminar | Fall 2019 - present |
| 3. | eCHT reading seminar on <i>Hopf algebras, rings, and algebroids</i>
Online. | Spring 2024 |
| 4. | eCHT reading seminar on <i>the Adams spectral sequence for tmf</i>
Online. | Spring 2023 |
| 5. | eCHT Kan Seminar
Online. | Spring 2022
Fall 2020 |
| 6. | Seminar on <i>Reflection positivity and invertible phases</i>
Online. | Spring 2022 |
| 7. | Seminar on <i>stable homotopy theory</i>
Online. | Spring 2021 |